

Blast Furnace Operations Manual

Purpose: The blast furnace is used for the production of molten iron from iron ore.

Normal Operating Parameters:

Temperature: 1450°C - 1550°C

Pressure: 1.5 - 2.0 bar

Hot Blast Temperature: 950°C - 1200°C

Inspection Checklist:

- Check cooling water flow.
- Verify blower operation.
- Monitor tuyere temperatures.
- Inspect refractory condition.
- Verify pressure readings.

Common Failure Modes:

- Refractory wear
- Cooling water leakage
- Blower malfunction
- Temperature instability

Maintenance Frequency:

Daily visual inspection. Monthly equipment inspection. Quarterly preventive maintenance.